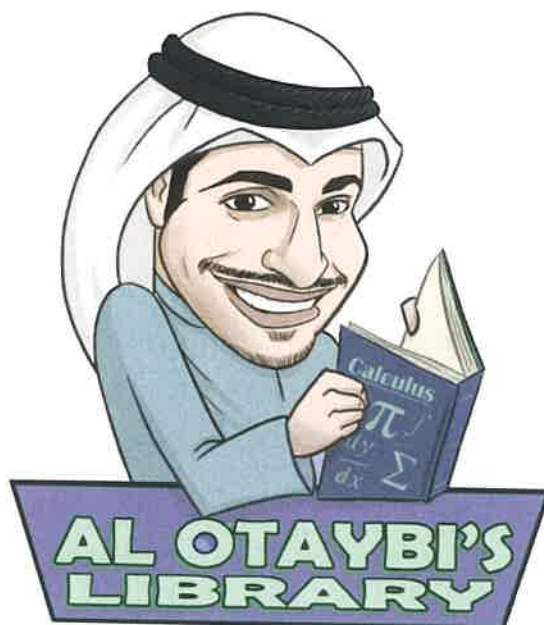


Kuwait University

2021/2022

Fall course

Fall course



Calculus A

First

مراجعة

اختبارات سابقة

( مراجعة الفيرست )

Eng. Mohammad Alotaybi 90048657 Twitter: @M\_Alodaybi



KUWAIT UNIVERSITY  
Faculty of Science  
Department of Mathematics

# Math 101: Calculus I

First Exam, 22 March 2018

Duration: 90 minutes

- In questions 1-9, show all your work.
- Calculators and communication devices are not allowed in the examination room.

Name: \_\_\_\_\_

Student ID: \_\_\_\_\_

Section Number: \_\_\_\_\_ Serial Number: \_\_\_\_\_

|             |  |
|-------------|--|
| Q1 (10 pts) |  |
| Q2 (10 pts) |  |
| Q3 (10 pts) |  |
| Q4 (10 pts) |  |
| Q5 (10 pts) |  |
| Q6 (10 pts) |  |
| Q7 (10 pts) |  |
| Q8 (10 pts) |  |
| Q9 (20 pts) |  |
| Total       |  |

1.  $[2.5 \times 4 = 10 \text{ pts.}]$  Use the given graph of  $f$  to evaluate each of the following limits, if it exists.

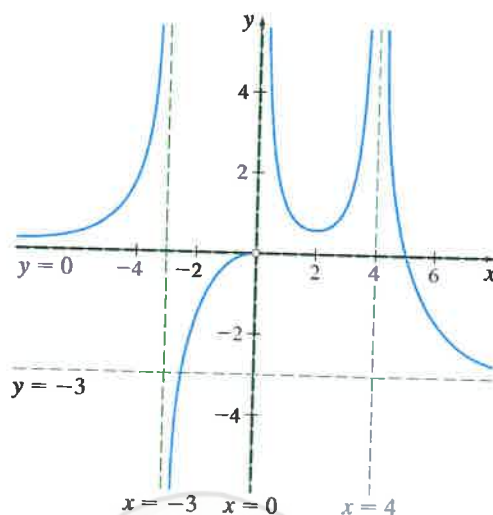


Figure 1: The graph of  $y = f(x)$ .

- (a)  $\lim_{x \rightarrow -3} f(x) =$
- (b)  $\lim_{x \rightarrow 0^-} f(x) =$
- (c)  $\lim_{x \rightarrow 0^+} f(x) =$
- (d)  $\lim_{x \rightarrow -\infty} f(x) =$
2.  $[5 + 5 = 10 \text{ pts.}]$  The graph of  $f$  is given below.

- (a) Explain why  $f$  is one-to-one.
- (b) Find  $f^{-1}(2)$ .

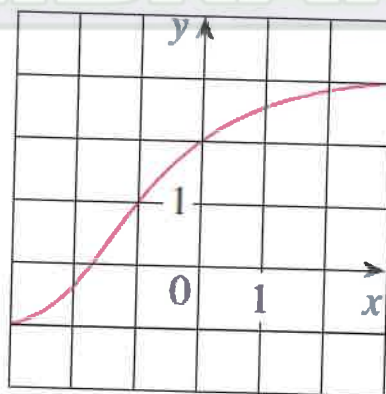


Figure 2: The graph of  $y = f(x)$ .

3. [10 pts.] Let  $f(x) = \ln(2^x - 1)$ . Find  $f^{-1}(x)$ .

4. [2 + 8 = 10 pts.] Define  $f(x) = \tan^{-1}\left(\frac{1}{x-2}\right)$  for  $x \neq 2$ .

(a) Explain why  $f$  is discontinuous at  $x = 2$ .

(b) Classify the type of this discontinuity.

5. [10 pts.] Find a value for the constant  $A$ , if any, that makes the function  $f$  continuous at  $x = 1$ , where

$$f(x) = \begin{cases} \frac{\sqrt{x+1} - \sqrt{2x}}{x-1} & \text{if } x \geq 0, x \neq 1 \\ A & \text{if } x = 1. \end{cases}$$

6. [10 pts.] Suppose that the function  $f$  is continuous on  $[0, 1]$  and satisfies  $0 < f(x) < 1$  for all  $x$  in  $[0, 1]$ . Use the Intermediate Value Theorem to show that the equation  $f(x) - x = 0$  has a solution in  $(0, 1)$ .

7. [10 pts.] Let  $f(x) = \frac{x^2 - 1}{e^x}$ . Find  $f'(1)$ .

3.2

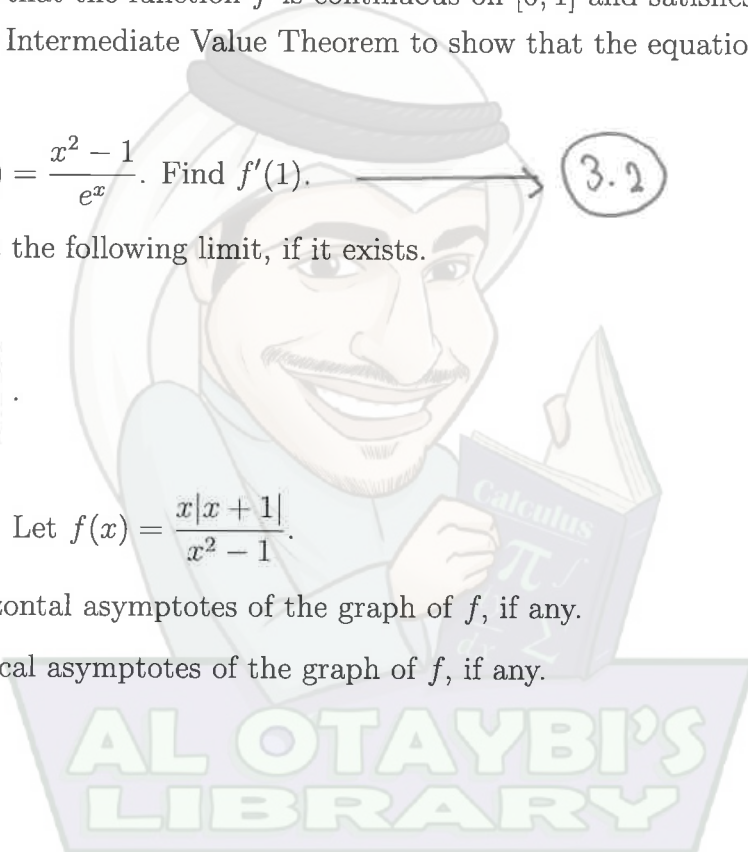
8. [10 pts.] Evaluate the following limit, if it exists.

$$\lim_{x \rightarrow 0} \left( \frac{\frac{1}{x^2 + 4} - \frac{1}{4}}{x} \right).$$

9. [10+10 = 20 pts.] Let  $f(x) = \frac{x|x+1|}{x^2 - 1}$ .

(a) Find the horizontal asymptotes of the graph of  $f$ , if any.

(b) Find the vertical asymptotes of the graph of  $f$ , if any.



1. [2.5 × 4 = 10 pts.] Use the given graph of  $f$  to evaluate each of the following limits, if it exists.

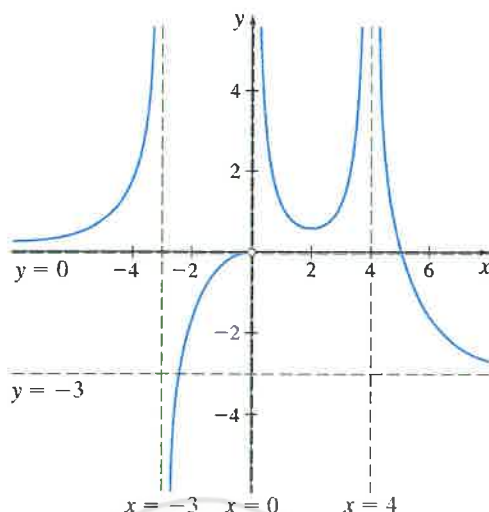


Figure 1: The graph of  $y = f(x)$ .

يفرق الاتجاهين : -3, 0, 4  
يفرق الصورة :

- (a)  $\lim_{x \rightarrow -3} f(x) = \text{DNE}$   $(\because \lim_{x \rightarrow -3^+} f(x) = +\infty \neq \lim_{x \rightarrow -3^-} f(x) = -\infty)$
- (b)  $\lim_{x \rightarrow 0^-} f(x) = 0$
- (c)  $\lim_{x \rightarrow 0^+} f(x) = +\infty$
- (d)  $\lim_{x \rightarrow -\infty} f(x) = 0$

2. [5 + 5 = 10 pts.] The graph of  $f$  is given below.

(a) Explain why  $f$  is one-to-one.

(b) Find  $f^{-1}(2)$ .

$\because$  No horizontal line intersect the graph more than one time  $\rightarrow \therefore f$  is 1-1.

$$\therefore f(0) = 2$$

$$\therefore f^{-1}(2) = 0$$

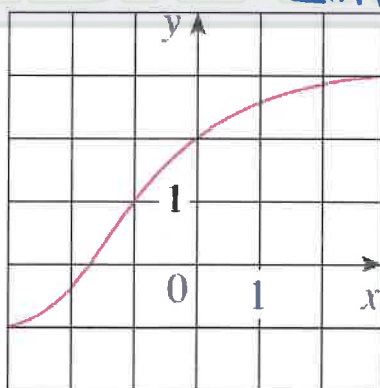


Figure 2: The graph of  $y = f(x)$ .

Q#3 Let  $F(x) = \ln(2^x - 1)$  Find  $F^{-1}(x)$

$$y = \ln(2^x - 1)$$

$$e^y = \cancel{e^{\ln(2^x - 1)}}$$

$$e^y = 2^x - 1$$

$$e^y + 1 = 2^x$$

$$\log_2(e^y + 1) = \cancel{\log_2 2^x}$$

$$x = \log_2(e^y + 1)$$

$$\therefore f^{-1}(x) = \log_2(e^x + 1)$$

Q#4 Define  $f(x) = \tan^{-1}\left(\frac{1}{x-2}\right)$  for  $x \neq 2$

a) Explain why  $f$  is discontinuous at  $x=2$ .

b) Classify the type of discontinuity

$f(2)$  undefined

$$\lim_{x \rightarrow 2} \tan^{-1}\left(\frac{1}{x-2}\right) = \tan^{-1}\left(\frac{1}{0}\right) = \tan^{-1}(\infty)$$

$$\begin{aligned} \lim_{x \rightarrow 2^+} \tan^{-1}\left(\frac{1}{x-2}\right) &= \tan^{-1}\left(\frac{1}{0^+}\right) = \tan^{-1}(+\infty) = +\frac{\pi}{2} \\ \lim_{x \rightarrow 2^-} \tan^{-1}\left(\frac{1}{x-2}\right) &= \tan^{-1}\left(\frac{1}{0^-}\right) = \tan^{-1}(-\infty) = -\frac{\pi}{2} \end{aligned}$$

$$\therefore \lim_{x \rightarrow 2^+} f(x) \neq \lim_{x \rightarrow 2^-} f(x)$$

$$\therefore \lim_{x \rightarrow 2} f(x) \text{ DNE}$$

$\therefore$  at  $x=2$   $f$  is not continuous

$$b) \therefore \lim_{x \rightarrow 2^+} f(x) \neq \lim_{x \rightarrow 2^-} f(x)$$

$\therefore$  at  $x=2 \rightarrow$  Jump discontinuity

Q#5

Find A

so that  $f$  is continuous at  $x=1$

$$f(x) = \begin{cases} \frac{\sqrt{x+1} - \sqrt{2x}}{x-1}, & x \neq 1 \\ A, & x = 1 \end{cases}$$

$$f(1) = \boxed{A}$$

$$= \lim_{x \rightarrow 1} \frac{\sqrt{x+1} - \sqrt{2x}}{x-1} \cdot \frac{\sqrt{x+1} + \sqrt{2x}}{\sqrt{x+1} + \sqrt{2x}}$$

$$= \lim_{x \rightarrow 1} \frac{x+1 - 2x}{(x-1) [\sqrt{x+1} + \sqrt{2x}]}$$

$$= \lim_{x \rightarrow 1} \frac{1 - x}{(x-1) [\sqrt{x+1} + \sqrt{2x}]}$$

$$= \lim_{x \rightarrow 1} \frac{-\cancel{(x-1)}}{\cancel{(x-1)} [\sqrt{x+1} + \sqrt{2x}]}$$

$$= \frac{-1}{\sqrt{2} + \sqrt{2}} = \boxed{\frac{-1}{2\sqrt{2}}}$$

∴  $f$  is continuous at  $x=1$

$$\therefore f(1) = \lim_{x \rightarrow 1} f(x)$$

$$\therefore A = \frac{-1}{2\sqrt{2}}$$

Q#6

$f$  is continuous on  $[0,1]$  ,  $0 \leq f(x) \leq 1$

Show that  $f(x) - x = 0$

has a solution in  $(0,1)$

Let  $h(x) = f(x) - x$

الافتقار

1)  $h(0) = f(0) - 0 = f(0) > 0$  since  $0 \leq f(x)$

2)  $h(1) = f(1) - 1 < 0$  since  $f(x) \leq 1$

From ①②  $\longrightarrow$  By I.V.T

Since  $h(x)$  is continuous on  $[0,1]$

then there exist at least one  $c \in (0,1)$

such that  $h(c) = 0$

$$f(c) - c = 0$$

$$f(c) = c$$

Q#7 Let  $f(x) = \frac{x^2 - 1}{e^x}$ . Find  $f'(1)$ .

$$f'(x) = \frac{e^x (2x) - (x^2 - 1) e^x}{(e^x)^2}$$

at  $x=1$   $f'(1) = \frac{e'(2) - 0}{(e')^2} = \frac{2e}{e^2} = \frac{2}{e}$

Q#8

$$\lim_{x \rightarrow 0} \left( \frac{\frac{1}{x^2 + 4} - \frac{1}{4}}{x} \right) \quad \infty$$

$$= \lim_{x \rightarrow 0} \frac{\frac{4}{4(x^2 + 4)} - \frac{x^2 + 4}{4(x^2 + 4)}}{x}$$

$$= \lim_{x \rightarrow 0} \frac{4 - (x^2 + 4)}{4(x^2 + 4) \cdot x}$$

$$= \lim_{x \rightarrow 0} \frac{\cancel{4} - x^2 - \cancel{4}}{4(x^2 + 4) \cdot x}$$

$$= \lim_{x \rightarrow 0} \frac{-x^2}{4(x^2 + 4) \cdot x} = \frac{0}{4(4)} = 0$$

Q#9

$$f(x) = \frac{x(x+1)}{x^2-1}$$

### Horizontal Asymptote

$$\lim_{x \rightarrow \infty} \frac{x(x+1)}{x^2-1}$$

$$\lim_{x \rightarrow \infty} \frac{\cancel{x} \cdot \cancel{x} (1 + \frac{1}{x})}{\cancel{x^2} (1 - \frac{1}{x^2})}$$

$$= \frac{1+0}{1-0} = \boxed{1}$$

$$\lim_{x \rightarrow -\infty} \frac{-x(x+1)}{x^2-1}$$

$$\lim_{x \rightarrow -\infty} \frac{-\cancel{x} \cdot \cancel{x} (1 + \frac{1}{x})}{\cancel{x^2} (1 - \frac{1}{x^2})}$$

$$= \frac{-(1+0)}{1-0} = \boxed{-1}$$

$\Rightarrow \therefore y=1$  and  $y=-1$  are HA's.

### Vertical Asymptote

at  $x=1$

$$\lim_{x \rightarrow 1^+} \frac{x(x+1)}{(x-1)(x+1)}$$

$$= \frac{1}{0^+} = +\infty$$

$$\lim_{x \rightarrow 1^-} \frac{x(x+1)}{(x-1)(x+1)}$$

$$= \frac{1}{0^-} = -\infty$$

$\Rightarrow \therefore x=1$  is V.A

at  $x=-1$

$$\lim_{x \rightarrow -1^+} \frac{+x(x+1)}{(x-1)(x+1)}$$

$$= \frac{-1}{-2} = \frac{1}{2}$$

$$\lim_{x \rightarrow -1^-} \frac{-x(x+1)}{(x-1)(x+1)}$$

$$= \frac{-1}{-2} = \frac{1}{2}$$

$\therefore x=-1$  is not V.A



KUWAIT UNIVERSITY  
Faculty of Science  
Department of Mathematics

# Math 101: Calculus I

First Exam, 22 March 2018

Duration: 90 minutes

- In questions 1-9, show all your work.
- Calculators and communication devices are not allowed in the examination room.

Name: \_\_\_\_\_

Student ID: \_\_\_\_\_

Section Number: \_\_\_\_\_ Serial Number: \_\_\_\_\_

|             |  |
|-------------|--|
| Q1 (10 pts) |  |
| Q2 (10 pts) |  |
| Q3 (10 pts) |  |
| Q4 (10 pts) |  |
| Q5 (10 pts) |  |
| Q6 (10 pts) |  |
| Q7 (10 pts) |  |
| Q8 (10 pts) |  |
| Q9 (20 pts) |  |
| Total       |  |

Calculators and communication devices are not allowed in the examination room.

**Answer all questions and justify all your answers.**

1. [10+10 = 20 pts.] Use the graph of  $y = f(x)$  shown below to:

a) Find the following limits:

i)  $\lim_{x \rightarrow -1^+} f(x)$ ;      ii)  $\lim_{x \rightarrow 0^-} f(x)$ ;      iii)  $\lim_{x \rightarrow 0^+} f(x)$ ;      iv)  $\lim_{x \rightarrow 1^+} f(x)$ ;      v)  $\lim_{x \rightarrow \infty} f(x)$ .

b) Find and classify the discontinuities of  $f$  as removable, infinite or jump.

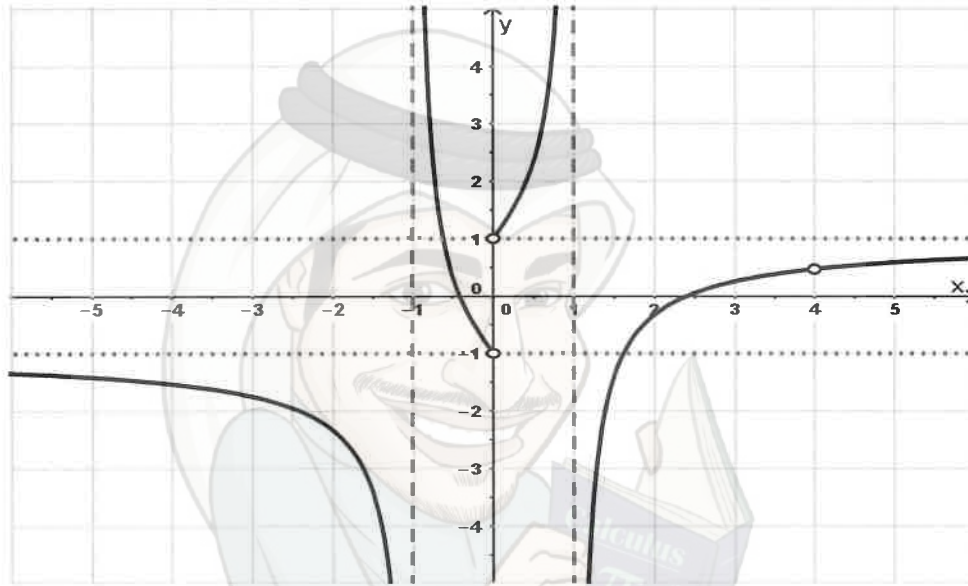


Figure 1: The graph of  $y = f(x)$ .

2. [10+10 = 20 pts.] Evaluate each of the following limits, if it exists.

a)  $\lim_{x \rightarrow 0} (x - 2)^2 \sin\left(\frac{\pi}{x - 2}\right)$ ;      b)  $\lim_{x \rightarrow 2} \left(\frac{1}{x - 2} - \frac{4}{x^2 - 4}\right)$ .

3. [10+10 = 20 pts.] Let  $f(x) = \frac{\sqrt{x^2 + 5}}{2x + 3}$ . Find the horizontal and vertical asymptotes of the graph of  $f$ .

4. [5+5 = 10 pts.] Let  $f(x) = 3^x + 1$ .

a) Find  $f^{-1}(x)$ .

b) Find the range of  $f$ .

5. [10 pts.] Determine whether  $f$  is continuous at  $x = 1$ , where: 
$$f(x) = \begin{cases} 2^x, & \text{if } x \geq 1 \\ \frac{1 - x^2}{|1 - x|}, & \text{if } x < 1. \end{cases}$$

6. [10 pts.] Let  $f(x) = \sqrt{x - 1}$ . Use the definition of the derivative to find  $f'(5)$ .

7. [10 pts.] Find the points on the graph:  $y = \frac{x}{x^2 + 1}$  where the tangent line is horizontal.

Q#1

$$\lim_{x \rightarrow -1^+} f(x) = \infty$$

$$\lim_{x \rightarrow 0^-} f(x) = -1$$

$$\lim_{x \rightarrow 0^+} f(x) = 1$$

$$\lim_{x \rightarrow 1^+} f(x) = -\infty$$

$$\lim_{x \rightarrow \infty} f(x) = 1$$

يفرق الـ ٤ :  $-1, 0, 1$

يفرق الصورة : 4

b) Find and classify the discontinuity point

at  $x = 4 \rightarrow$  removable disc. ( $f(4) \neq \lim_{x \rightarrow 4} f(x)$ )

at  $x = -1 \rightarrow$  infinite disc. ( $\lim_{x \rightarrow -1^+} f(x) = \infty$   
 $\lim_{x \rightarrow -1^-} f(x) = -\infty$ )

at  $x = 0 \rightarrow$  Jump disc. ( $\lim_{x \rightarrow 0^+} f(x) \neq \lim_{x \rightarrow 0^-} f(x)$ )

at  $x = 1 \rightarrow$  infinite disc. ( $\lim_{x \rightarrow 1^+} f(x) = -\infty$   
 $\lim_{x \rightarrow 1^-} f(x) = \infty$ )

Q# 2

a)  $\lim_{x \rightarrow 0} (x-2)^2 \sin\left(\frac{\pi}{x-2}\right)$

$$= (0-2)^2 \sin \frac{\pi}{0-2} = (-2)^2 \sin\left(\frac{-\pi}{2}\right)$$

$$= 4 \left( -\sin \frac{\pi}{2} \right) = -4 (1) = -4$$

---

b)  $\lim_{x \rightarrow 2} \left( \frac{1}{x-2} - \frac{4}{x^2-4} \right)$

$\infty - \infty$

$$= \lim_{x \rightarrow 2} \left( \frac{1}{x-2} - \frac{4}{(x-2)(x+2)} \right)$$

$$= \lim_{x \rightarrow 2} \left( \frac{x+2}{(x-2)(x+2)} - \frac{4}{(x-2)(x+2)} \right)$$

$$= \lim_{x \rightarrow 2} \frac{x+2-4}{(x-2)(x+2)}$$

$$= \lim_{x \rightarrow 2} \frac{\cancel{x}-2}{(\cancel{x}-2)(x+2)}$$

$$= \frac{1}{4}$$

Q#13

$$\text{Let } f(x) = \frac{\sqrt{x^2 + 5}}{2x + 3}$$

HA?  
V.A?Horizontal Asymptote

$$\lim_{x \rightarrow \pm\infty} \frac{\sqrt{x^2 + 5}}{2x + 3} \quad \frac{\infty}{\pm\infty}$$

$$\lim_{x \rightarrow \pm\infty} \frac{\sqrt{x^2(1 + \frac{5}{x^2})}}{2x + 3} = \lim_{x \rightarrow \pm\infty} \frac{|x| \sqrt{1 + \frac{5}{x^2}}}{2x + 3}$$

$$\lim_{x \rightarrow \infty} \frac{x \sqrt{1 + \frac{5}{x^2}}}{2x + 3}$$

$$\lim_{x \rightarrow -\infty} \frac{-x \sqrt{1 + \frac{5}{x^2}}}{2x + 3}$$

$$\lim_{x \rightarrow \infty} \frac{\cancel{x} \sqrt{1 + \frac{5}{x^2}}}{\cancel{x} [2 + \frac{3}{x}]}$$

$$\lim_{x \rightarrow -\infty} \frac{-\cancel{x} \sqrt{1 + \frac{5}{x^2}}}{\cancel{x} [2 + \frac{3}{x}]}$$

$$= \frac{\sqrt{1+0}}{2+0} = \frac{1}{2}$$

$$= \frac{-\sqrt{1+0}}{2+0} = -\frac{1}{2}$$

∴  $y = \frac{1}{2}$  and  $y = -\frac{1}{2}$  are HA's

Vertical Asymptote

$$2x + 3 = 0 \rightarrow 2x = -3$$

$$x = -\frac{3}{2}$$

$$\lim_{x \rightarrow -\frac{3}{2}^+} \frac{\sqrt{x^2 + 5}}{2x + 3}$$

$$\lim_{x \rightarrow -\frac{3}{2}^-} \frac{\sqrt{x^2 + 5}}{2x + 3}$$

$$= \frac{\sqrt{9/4 + 5}}{0^+} = +\infty$$

$$= \frac{\sqrt{9/4 + 5}}{0^-} = -\infty$$

∴  $x = -\frac{3}{2}$  is V.A

Q# 4       $f(x) = 3^x + 1$

- a)  $f^{-1}$   
b) range of  $f$

a)  $y = 3^x + 1$

$y - 1 = 3^x$

$\log_3(y-1) = \log_3 3^x$

$x = \log_3(y-1)$

$f^{-1}(x) = \log_3(x-1)$

b) Domain of  $f^{-1}(x)$  :  $x - 1 > 0$   
 $x > 1$   
 $(1, \infty)$

$\therefore$  Range of  $f = (1, \infty)$

Q#5  $f(x) = \begin{cases} 2^x, & x \geq 1 \\ \frac{1-x^2}{|1-x|}, & x < 1 \end{cases}$

Continuous or not at  $x=1$

$$f(1) = 2^1$$

$$\lim_{x \rightarrow 1^+} 2^x = 2^1$$

$$\lim_{x \rightarrow 1^-} \frac{1-x^2}{|1-x|}$$

$$\lim_{x \rightarrow 1^-} \frac{1-x^2}{+(1-x)}$$

$$\lim_{x \rightarrow 1^-} \frac{(1-x)(1+x)}{1-x} = 2$$

$$\therefore f(1) = \lim_{x \rightarrow 1^+} f(x) = \lim_{x \rightarrow 1^-} f(x) = 2$$

$\therefore$  at  $x=1$   $f$  is continuous.

Q#6

$$f(x) = \sqrt{x-1}$$

Use definition  
to find  $f'(5)$

$$f'(a) = \lim_{x \rightarrow a} \frac{f(x) - f(a)}{x - a}$$

$$f'(5) = \lim_{x \rightarrow 5} \frac{\sqrt{x-1} - f(5)}{x - 5} \quad \text{with } f(5) = \sqrt{4} = 2$$

$$= \lim_{x \rightarrow 5} \frac{\sqrt{x-1} - 2}{x - 5} \times \frac{\sqrt{x-1} + 2}{\sqrt{x-1} + 2}$$

$$= \lim_{x \rightarrow 5} \frac{x-1-4}{(x-5)(\sqrt{x-1}+2)}$$

$$= \lim_{x \rightarrow 5} \frac{x-5}{(x-5)(\sqrt{x-1}+2)}$$

$$= \frac{1}{\sqrt{4} + 2}$$

$$= \frac{1}{2+2} = \frac{1}{4}$$

Q#7

$$y = \frac{x}{x^2 + 1}$$

Horizontal tangent line?

$$y' = \frac{(x^2 + 1)(1) - x(2x)}{(x^2 + 1)^2}$$

$$= \frac{x^2 + 1 - 2x^2}{(x^2 + 1)^2} = \frac{1 - x^2}{(x^2 + 1)^2}$$

∴ Tangent line is horizontal

$$∴ y' = 0$$

$$∴ 1 - x^2 = 0 \rightarrow x^2 = 1 \begin{cases} x = -1 \\ x = 1 \end{cases}$$

$$\text{at } x = -1 \rightarrow y = \frac{-1}{1 + 1} = -\frac{1}{2}$$

$$\text{at } x = 1 \rightarrow y = \frac{1}{1 + 1} = \frac{1}{2}$$

$$∴ \left(-1, -\frac{1}{2}\right), \left(1, \frac{1}{2}\right)$$



KUWAIT UNIVERSITY  
Faculty of Science  
Department of Mathematics

## Math 101: Calculus I

First Exam, 07 March 2019

Duration: 90 minutes

- In questions 1-8, show all your work.
- Calculators and communication devices are not allowed in the examination room.

Name: \_\_\_\_\_

Student ID: \_\_\_\_\_

Section Number: \_\_\_\_\_ Serial Number: \_\_\_\_\_

|              |  |
|--------------|--|
| Q1 (25 pts)  |  |
| Q2 (15 pts)  |  |
| Q3 (5 pts)   |  |
| Q4 (10 pts)  |  |
| Q5 (10 pts)  |  |
| Q6 (10 pts)  |  |
| Q7 (10 pts)  |  |
| Q8 (15 pts)  |  |
| <b>Total</b> |  |

1.  $[3 + 3 + 3 + 3 + 3 + 5 + 5 = 25 \text{ pts.}]$  Let

$$f(x) = \begin{cases} e^x - \frac{3}{x-7} + 5, & \text{if } x \leq 1, \\ \ln(x-1) + 3, & \text{if } 1 < x \leq 3 \\ \frac{x^2+1}{x^3+1}, & \text{if } x > 3. \end{cases}$$

(a) Find  $\lim_{x \rightarrow 1} f(x)$ , if it exists.

$$\lim_{x \rightarrow 1^+} (\ln(x-1) + 3) = \ln 0 + 3 = -\infty, \quad \lim_{x \rightarrow 1^-} (e^x - \frac{3}{x-7} + 5) = e^1 - \frac{3}{-6} + 5$$

$$\Rightarrow \because \lim_{x \rightarrow 1^+} f(x) \neq \lim_{x \rightarrow 1^-} f(x) \Rightarrow \therefore \lim_{x \rightarrow 1} f(x) \text{ DNE}$$

(b) Find  $\lim_{x \rightarrow 3} f(x)$ , if it exists.

$$\lim_{x \rightarrow 3^+} \frac{x^2+1}{x^3+1} = \frac{9+1}{27+1} = \frac{10}{28} = \frac{5}{14}, \quad \lim_{x \rightarrow 3^-} (\ln(x-1) + 3) = \ln 2 + 3$$

$$\Rightarrow \because \lim_{x \rightarrow 3^+} f(x) \neq \lim_{x \rightarrow 3^-} f(x) \Rightarrow \therefore \lim_{x \rightarrow 3} f(x) \text{ DNE}$$

(c) Find  $\lim_{x \rightarrow -\infty} f(x)$ , if it exists.

$$\lim_{x \rightarrow -\infty} (e^x - \frac{3}{x-7} + 5) = e^{-\infty} - \frac{3}{-\infty} + 5 = 0 - 0 + 5 = 5$$

(d) Find  $\lim_{x \rightarrow \infty} f(x)$ , if it exists.

$$\lim_{x \rightarrow \infty} \frac{x^2+1}{x^3+1} \stackrel{\infty}{=} \lim_{x \rightarrow \infty} \frac{x^2(1+\frac{1}{x^2})}{x^3(1+\frac{1}{x^3})} = \frac{1+0}{\infty(1+0)} = 0$$

(e) Find, if any, all vertical asymptotes of the graph of  $f$ .

$$x=1 \rightarrow \text{because } \lim_{x \rightarrow 1^+} f(x) = -\infty$$

(f) Find and classify, if any, all the points of discontinuity.

$$x=1 \rightarrow \text{infinite disc.}$$

$$x=3 \rightarrow \text{Jump disc.}$$

(g) Find, if any, all horizontal asymptotes of the graph of  $f$ .

$$y=0$$

$$y=5$$

2. [5 + 5 + 5 = 15 pts.] Let  $f(x) = \frac{x+1}{2x+1}$ .

\*Trick\* (a) Show that  $f$  is one-to-one.

(b) Find the inverse function of  $f$ .

(c) Find the domain and range of  $f$ .

(a) Video

$$\Rightarrow f(x_1) = f(x_2)$$

$$\frac{x_1+1}{2x_1+1} = \frac{x_2+1}{2x_2+1}$$

$$(x_1+1)(2x_2+1) = (x_2+1)(2x_1+1)$$

$$2x_1x_2 + x_1 + 2x_2 + 1 = 2x_1x_2 + x_2 + 2x_1 + 1$$

$$\Rightarrow \text{the } \boxed{x_1 = x_2} \Rightarrow \therefore f \text{ is 1-1}$$

(b)

$$y = \frac{x+1}{2x+1}$$

$$y(2x+1) = x+1$$

$$2xy + y = x+1$$

$$2xy - x = 1-y$$

$$x(2y-1) = 1-y$$

$$x = \frac{1-y}{2y-1}$$

$$f^{-1}(x) = \frac{1-x}{2x-1}$$

(c)

$$\text{Domain of } f(x): \begin{aligned} 2x+1 &= 0 \\ 2x &= -1 \\ x &= -\frac{1}{2} \end{aligned}$$

$$\Rightarrow \text{Domain of } f(x) = \mathbb{R} \setminus \{-\frac{1}{2}\}$$

$$\text{Domain of } f^{-1}(x): \begin{aligned} 2x-1 &= 0 \\ 2x &= 1 \\ x &= \frac{1}{2} \end{aligned}$$

$$\Rightarrow \text{Domain of } f^{-1}(x) = \mathbb{R} \setminus \{\frac{1}{2}\}$$

$$D_f = \mathbb{R} \setminus \{-\frac{1}{2}\} \rightarrow \text{Range of } f^{-1} = \mathbb{R} \setminus \{-\frac{1}{2}\}$$

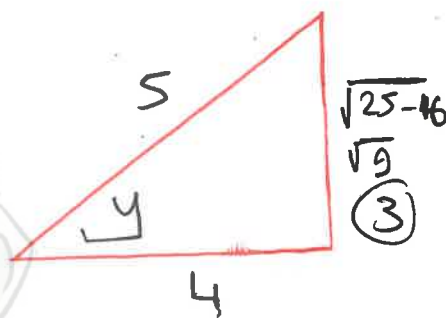
$$D_{f^{-1}} = \mathbb{R} \setminus \{\frac{1}{2}\} \rightarrow \text{Range of } f = \mathbb{R} \setminus \{\frac{1}{2}\}$$

3. [5 pts.] Find the exact value of  $\sin \left( \cos^{-1} \left( \frac{4}{5} \right) \right)$ .

$$y = \cos^{-1} \left( \frac{4}{5} \right)$$

$$\cos y = \cos \cos^{-1} \left( \frac{4}{5} \right)$$

$$\cos y = \frac{4}{5}$$



$$\sin \left( \cos^{-1} \left( \frac{4}{5} \right) \right)$$

$$\sin y = \frac{3}{5}$$

AL OTAYBI'S  
LIBRARY

4. [10 pts.] Solve the equation  $\ln(x^2 - 2x - 2) = 0$ .

$$e^{\ln(x^2 - 2x - 2)} = e^0$$

$$x^2 - 2x - 2 = 1$$

$$x^2 - 2x - 3 = 0$$

$$(x + 1)(x - 3) = 0$$

$$- \quad x + 1 = 0 \quad x - 3 = 0$$

$$\Rightarrow x = -1, \quad x = 3$$

5. [10 pts.] If  $e^x \leq f(x) \leq \frac{4x^2 + 1}{x^2}$ , find  $\lim_{x \rightarrow 0} x^4 f(x)$ , if it exists.

$$\because x^4 > 0$$

$$x^4 e^x \leq x^4 f(x) \leq x^4 \frac{4x^2 + 1}{x^2}$$

$$x^4 e^x \leq x^4 f(x) \leq x^2 (4x^2 + 1)$$

$$\lim_{x \rightarrow 0} x^4 e^x = 0$$

$$\lim_{x \rightarrow 0} x^2 (4x^2 + 1) = 0$$

$\Rightarrow \therefore$  Sandwich Thm (Squeeze Thm)

$$\lim_{x \rightarrow 0} x^4 f(x) = 0$$

6. [10 pts.] Find the values of  $a$  and  $b$ , if any, that make  $f$  continuous everywhere.

$$f(x) = \begin{cases} \frac{x^2 - 4}{x - 2}, & \text{if } x < 2, \\ ax^2 - bx + 4, & \text{if } 2 \leq x < 3 \\ 2x - a + b, & \text{if } x \geq 3. \end{cases}$$

$$f(2) = 4a - 2b + 4$$

$$\lim_{x \rightarrow 2^+} (ax^2 - bx + 4) = 4a - 2b + 4$$

$$\lim_{x \rightarrow 2^-} \frac{x^2 - 4}{x - 2} = \lim_{x \rightarrow 2^-} \frac{(x-2)(x+2)}{x-2} = 4$$

$\therefore f$  is continuous everywhere

$$\therefore f(2) = \lim_{x \rightarrow 2^+} f(x) = \lim_{x \rightarrow 2^-} f(x)$$

$$4a - 2b + 4 = 4a - 2b + 4 = 4$$

$$4a - 2b + 4 = 4$$

$$4a - 2b = 0$$

$$\rightarrow 2b = 4a$$

$$\div 2$$

$$b = 2a$$

①

$$f(3) = 6 - a + b$$

$$\lim_{x \rightarrow 3^+} (2x - a + b) = 6 - a + b$$

$$\lim_{x \rightarrow 3^-} (ax^2 - bx + 4) = 9a - 3b + 4$$

$\therefore f$  is continuous everywhere

$$\therefore f(3) = \lim_{x \rightarrow 3^+} f(x) = \lim_{x \rightarrow 3^-} f(x)$$

$$6 - a + b = 6 - a + b = 9a - 3b + 4$$

$$6 - a + b = 9a - 3b + 4$$

$$6 - a + b - 9a + 3b - 4 = 0$$

$$2 - 10a + 4b = 0 \rightarrow \textcircled{2}$$

① in ②

$$2 - 10a + 4(2a) = 0$$

$$2 - 10a + 8a = 0$$

$$2 - 2a = 0$$

$$2a = 2$$

$$a = 1$$

in ①

$$b = 2$$

7. [10 pts.] Use the **definition of the derivative** to find  $f'(3)$  of  $f(x) = \sqrt{x-2} + 3$ .

$$f'(a) = \lim_{x \rightarrow a} \frac{f(x) - f(a)}{x - a}$$

$$= \lim_{x \rightarrow 3} \frac{(\sqrt{x-2} + 3) - f(3)}{x - 3} \rightarrow \begin{array}{r} \sqrt{3-2} + 3 \\ 1 + 3 \\ 1 + 3 \end{array}$$

$$= \lim_{x \rightarrow 3} \frac{\sqrt{x-2} + 3 - (1+3)}{x-3}$$

$$= \lim_{x \rightarrow 3} \frac{\sqrt{x-2} + \cancel{3} - 1 - \cancel{3}}{x-3}$$

$$= \lim_{x \rightarrow 3} \frac{\sqrt{x-2} - 1}{x-3} \quad \begin{array}{l} \text{g} \\ * \end{array} \frac{\sqrt{x-2} + 1}{\sqrt{x-2} + 1}$$

$$= \lim_{x \rightarrow 3} \frac{\sqrt{x-2} - 1}{(x-3)(\sqrt{x-2} + 1)}$$

$$= \lim_{x \rightarrow 3} \frac{\cancel{x-3}}{(\cancel{x-3})(\sqrt{x-2} + 1)} = \frac{1}{1+1} = \left(\frac{1}{2}\right)$$

8. [5 + 10 = 15 pts.] a) State the Intermediate Value Theorem.

b) Show that there is at least one real root of the equation  $\cos(\sqrt{x}) = e^x - 2$ .

a) If  $f(x)$  is continuous function on  $[a, b]$ , and  $w$  is a real number,  $w \in (f(a), f(b))$   
then there exist at least one  $c \in (a, b)$   
such that  $f(c) = w$

b)  $\cos \sqrt{x} = e^x - 2$

$$\cos \sqrt{x} - e^x + 2 = 0$$

$$\text{Let } f(x) = \cos \sqrt{x} - e^x + 2$$

$$\textcircled{1} f(0) = \cos 0 - e^0 + 2 = 1 - 1 + 2 = 2 > 0$$

$$\textcircled{2} f(4) = \cos \sqrt{4} - e^4 + 2$$
$$= \cos 2 - e^4 + 2 < 0$$

$-1 < \cos 2 < 1$        $\approx (2.7)^4$

From  $\textcircled{1}$   $\textcircled{2} \rightarrow \therefore$  By I.V.T

Since  $f(x)$  is continuous on  $[0, \infty)$  ( $\sqrt{x}$  Root function)

then  $f(x)$  is continuous on  $[0, 4]$

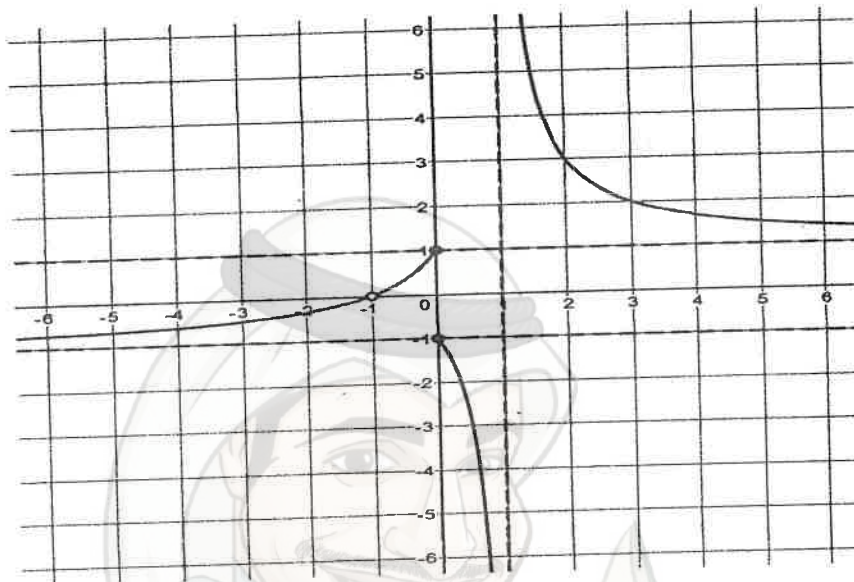
then there exist at least one  $c \in (0, 4)$

$$\text{s.t. } f(c) = 0$$

Calculators and communication devices are not allowed in the examination room.

Answer all questions and justify all your answers.

1.  $[5 \times 4 = 20 \text{ pts.}]$  Use the given graph of  $f$  to select the correct answer.



- (I)  $\lim_{x \rightarrow 0} f(x) =$   
(a) 1                      (b) 0                      (c)  $\infty$                       (d) -1                      (e) None of the mentioned

- (II)  $\lim_{x \rightarrow 1^+} f(x) =$   
(a)  $+\infty$                       (b)  $-\infty$                       (c) 1                      (d) 0                      (e) None of the mentioned

- (III) The graph of  $f$  has  
a) no vertical asymptotes  
b) only one vertical asymptote  
c) only one horizontal asymptote  
d) no horizontal asymptotes  
e) None of the above

- (IV) The function  $f$  has  
a) a jump discontinuity at  $x = 1$   
b) a removable discontinuity at  $x = 0$   
c) a jump discontinuity at  $x = -1$   
d) a removable discontinuity at  $x = -1$   
e) None of the above

2. [10 + 10 = 20 pts.] Evaluate each of the following limits, if it exists.

(a)  $\lim_{x \rightarrow \infty} (\sqrt{4x^2 + 1} - 2x) =$

(b)  $\lim_{x \rightarrow 1} \sin^{-1} \left( \frac{x-1}{x^2-1} \right) =$

3. [10 + 10 = 20 pts.] Let  $f(x) = \frac{x^2 - 4x}{|x|(x+2)}$ .

(a) Find and classify the discontinuities of  $f$  as removable, infinite or jump.

(b) Find the horizontal and vertical asymptotes of the graph of  $f$ , if any.

4. [10 pts.] Find the exact value of each expression

(a)  $\tan \left( \sin^{-1} \left( \frac{2}{3} \right) \right).$

(b)  $\log_{10} 4 + 2 \log_{10} 5.$

5. [10 pts.] Let  $f(x) = e^{x+1} + 2$ . Find  $f^{-1}(x)$ .

6. [10 pts.] Let  $f(x)$  be a function which satisfies  $5x - 6 \leq f(x) \leq x^2 + 3x - 5$  for all  $x \geq 0$ . Find  $\lim_{x \rightarrow 1} f(x)$ .

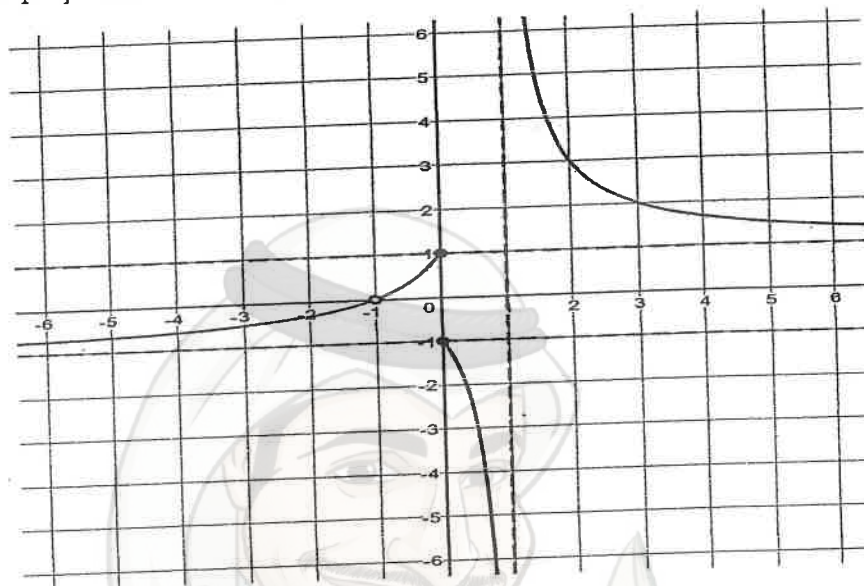
7. [10 pts.] Use the Intermediate Value Theorem to show that the equation  $e^x + \cos x = 4$  has a real root.



Calculators and communication devices are not allowed in the examination room.

Answer all questions and justify all your answers.

1.  $[5 \times 4 = 20 \text{ pts.}]$  Use the given graph of  $f$  to select the correct answer.



يفرق بالانجاس  
1, 0

(I)  $\lim_{x \rightarrow 0} f(x) = \longrightarrow$

DNE الكيد

(a) 1

(b) 0

(c)  $\infty$

(d) -1

(e) None of the mentioned

(II)  $\lim_{x \rightarrow 1^+} f(x) =$

(a)  $+\infty$

(b)  $-\infty$

(c) 1

(d) 0

(e) None of the mentioned

(III) The graph of  $f$  has

a) no vertical asymptotes

b) only one vertical asymptote

c) only one horizontal asymptote

d) no horizontal asymptotes

e) None of the above

$x = 0$

(IV) The function  $f$  has

a) a jump discontinuity at  $x = 1$

b) a removable discontinuity at  $x = 0$

c) a jump discontinuity at  $x = -1$

d) a removable discontinuity at  $x = -1$

e) None of the above

$x = -1 \rightarrow$  Removable

$x = 0 \rightarrow$  jump

$x = 1 \rightarrow$  infinite

Q #2

(a)  $\lim_{x \rightarrow \infty} (\sqrt{4x^2+1} - 2x) \quad \infty - \infty$

$$\lim_{x \rightarrow \infty} (\sqrt{4x^2+1} - 2x) \frac{\sqrt{4x^2+1} + 2x}{\sqrt{4x^2+1} + 2x}$$

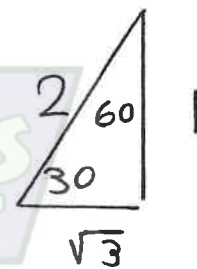
$$\lim_{x \rightarrow \infty} \frac{\cancel{4x^2+1} - \cancel{4x^2}}{\sqrt{4x^2+1} + 2x} = \frac{1}{\infty + \infty} = \frac{1}{\infty} = 0$$

(b)  $\lim_{x \rightarrow 1} \sin^{-1} \left( \frac{x-1}{x^2-1} \right) \quad \sin^{-1} \frac{0}{0}$

$$= \lim_{x \rightarrow 1} \sin^{-1} \left( \frac{\cancel{x-1}}{(\cancel{x-1})(x+1)} \right)$$

$$= \sin^{-1} \left( -\frac{1}{2} \right)$$

$$= \frac{\pi}{6}$$



Q#3

$$f(x) = \frac{x^2 - 4x}{|x| (x+2)}$$

(a)  $|x| (x+2) = 0$

$x=0$     $x=-2$

at  $x=0$

$f(0)$  undefined

$$\lim_{x \rightarrow 0^+} \frac{x^2 - 4x}{|x| (x+2)}$$

$\downarrow$   
 $0 + 0.1 = 0.1$

$$\lim_{x \rightarrow 0^+} \frac{x(x-4)}{x(x+2)}$$

$$= \frac{-4}{2} = -2$$

$$\lim_{x \rightarrow 0^-} \frac{x^2 - 4x}{|x| (x+2)}$$

$$\lim_{x \rightarrow 0^-} \frac{x(x-4)}{-x(x+2)}$$

$$= \frac{-4}{-(2)} = 2$$

$\therefore \lim_{x \rightarrow 0^+} f(x) \neq \lim_{x \rightarrow 0^-} f(x) \Rightarrow$  at  $x=0$  Jump disc

at  $x=-2$

$f(-2)$  undefined

$$\lim_{x \rightarrow -2} \frac{x^2 - 4x}{-|x| (x+2)}$$

$$= \frac{4+4}{0} = \frac{8}{0} = \infty$$

$\Rightarrow$

مطلوب

$$\lim_{x \rightarrow -2^+} \frac{x^2 - 4x}{-(x)(x+2)}$$

$-2 + 0.1 = -1.9$

$$= \frac{8}{0^+} = +\infty$$

$$\lim_{x \rightarrow -2^-} \frac{x^2 - 4x}{-(x)(x+2)}$$

$-2 - 0.1 = -2.1$

$$= \frac{8}{0^-} = -\infty$$

$\therefore$  at  $x = -2 \Rightarrow$  Infinite Disc.

(b) HA

$$\lim_{x \rightarrow \infty} \frac{x^2 - 4x}{x(x+2)}$$

$$\lim_{x \rightarrow \infty} \frac{x^2(1 - \frac{4}{x})}{x \cdot x(1 + \frac{2}{x})}$$

$$= \frac{1-0}{1+0} = 1$$

$$\lim_{x \rightarrow -\infty} \frac{x^2 - 4x}{-(x)(x+2)}$$

$$\lim_{x \rightarrow -\infty} \frac{x^2(1 - \frac{4}{x})}{-x \cdot x(1 + \frac{2}{x})}$$

$$= \frac{1-0}{-(1+0)} = -1$$

$y = 1$  and  $y = -1$  are HA's.

V.A

From (a)

$$\therefore \lim_{x \rightarrow -2^+} \frac{x^2 - 4x}{-x(x+2)} = +\infty$$

$$\lim_{x \rightarrow -2^-} \frac{x^2 - 4x}{-x(x+2)} = -\infty$$

$\therefore x = -2$  is V.A

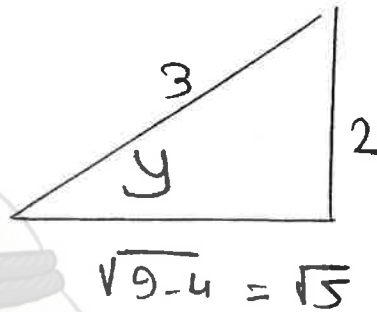
# Q#4

(a)  $\tan(\sin^{-1} \frac{2}{3})$

$$y = \sin^{-1} \frac{2}{3}$$

$$\sin y = \sin \sin^{-1} \frac{2}{3}$$

$$\sin y = \frac{2}{3}$$



→  $\tan(\sin^{-1} \frac{2}{3})$   
 $\tan y = \frac{2}{\sqrt{5}}$

(b)  $\log_{10} 4 + 2 \log_{10} 5$

$$\log_{10} 4 + \log_{10} 5^2$$

$$\log_{10} 4 + \log_{10} 25$$

$$\log_{10} (4 \cdot 25)$$

$$\log_{10} (100) = \log_{10} (10^2)$$

$$= 2 \quad \#$$

Q # 5

Let  $f(x) = e^{x+1} + 2$ . Find  $f^{-1}(x)$

$$y = e^{x+1} + 2$$

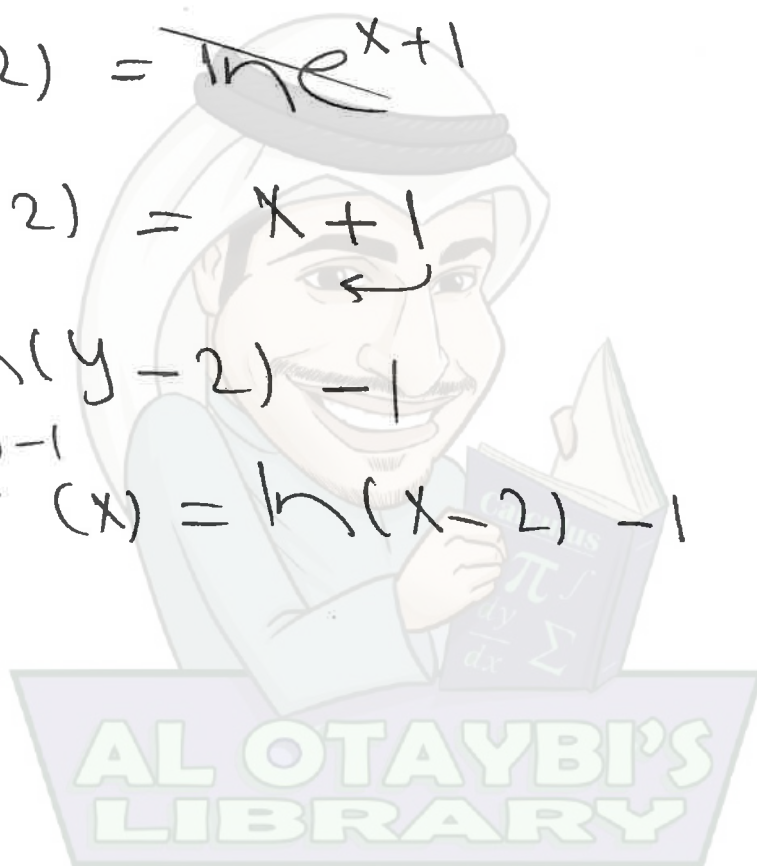
$$y - 2 = e^{x+1}$$

$$\ln(y-2) = \ln e^{x+1}$$

$$\ln(y-2) = x+1$$

$$x = \ln(y-2) - 1$$

$$\Rightarrow f^{-1}(x) = \ln(x-2) - 1$$



Q#6

$$5x - 6 \leq f(x) \leq x^2 + 3x - 5$$

Find  $\lim_{x \rightarrow 1} f(x)$

$$\lim_{x \rightarrow 1} (5x - 6) = 5 - 6 = \textcircled{-1}$$

$$\lim_{x \rightarrow 1} (x^2 + 3x - 5) = 1 + 3 - 5 = \textcircled{-1}$$

∴ By Sandwich Thm

$$\lim_{x \rightarrow 1} f(x) = -1$$

Q # 7

$$e^x + \cos x = 4$$

$$e^x + \cos x - 4 = 0$$

Let  $h(x) = e^x + \cos x - 4 \neq 0$

①  $h(0) = e^0 + \cos 0 - 4 = 1 + 1 - 4 = -2 < 0$

②  $h(2\pi) = e^{2\pi} + \cos 2\pi - 4$   
 $= e^{2\pi} + 1 - 4$   
 $= e^{2\pi} - 3 > 0$

From ① ②  $\rightarrow$  By I.V.T

Since  $h(x)$  is continuous on  $\mathbb{R}$  [  $e^x$ : exponential  
 $\cos x$ : Trigon.

then  $h(x)$  is continuous on  $[0, 2\pi]$

then these exist at least one  $c \in (0, 2\pi)$

s.t.  $h(c) = 0$

Kuwait University  
Department of Mathematics

Calculus I (Math 101)  
First Exam

5 November 2016  
Duration 90 minutes

Answer all questions. Calculators and mobile phones are not allowed.

1. (10 + 10 = 20 pts) Evaluate each of the following limits, if it exists.

(a)  $\lim_{x \rightarrow -2} \frac{3x^2 + 5x - 2}{x^2 - 3x - 10}.$

(b)  $\lim_{x \rightarrow -\infty} (x + \sqrt{x^2 + 2}).$

2. (20 pts) Let  $f(x) = \begin{cases} \frac{x^2 + x - 6}{x + 3}, & \text{if } x < 0 \\ \frac{x + 1}{(x - 1)^2}, & \text{if } x \geq 0. \end{cases}$

Find all the points of discontinuity of  $f$ . Classify each discontinuity as removable, jump, or infinite.

3. (10 + 5 = 15 pts) Let  $f(x) = \frac{2x^2 + 3}{x\sqrt{x^2 + 4}}.$

- (a) Find the horizontal asymptotes of the graph of  $f$ , if any.  
(b) Find the vertical asymptotes of the graph of  $f$ , if any.

4. (10 + 5 = 15 pts) Let  $f(x) = \ln(2x + 3).$

- (a) Find  $f^{-1}(x).$   
(b) Find the domain and the range of the function  $f.$

5. (10 pts) If  $(5 - 2|x - 4|) \leq f(x) + 2 \leq (x + 1)$  for all  $x$ , evaluate  $\lim_{x \rightarrow 4} f(x).$

6. (10 pts) Let  $f(x) = \sqrt{2x + 1}.$  Use the definition of the derivative to find  $f'(4).$

7. (5 + 5 = 10 pts)

(a) Show that  $\sin(\tan^{-1} x) = \frac{x}{\sqrt{1 + x^2}}.$

(b) Find the exact value of  $\sin\left(\tan^{-1}\left(\frac{1}{\sqrt{3}}\right)\right).$

Q #1

$$a) \quad \lim_{x \rightarrow -2} \frac{3x^2 + 5x - 2}{x^2 - 3x - 10} \quad \frac{3(4) - 10 - 2}{4 + 6 - 10}$$

$$x - (-2) = x + 2$$

$$\frac{0}{0}$$

$$\lim_{x \rightarrow -2} \frac{(3x - 1)(\cancel{x + 2})}{(\cancel{x + 2})(x - 5)}$$

$$= \frac{-6 - 1}{-7} = \frac{-7}{-7} = \textcircled{1} \quad \#$$

$$(b) \quad \lim_{x \rightarrow -\infty} (x + \sqrt{x^2 + 2}) \quad -\infty + \infty$$

$$\lim_{x \rightarrow -\infty} (x + \sqrt{x^2 + 2}) \quad \frac{x - \sqrt{x^2 + 2}}{x - \sqrt{x^2 + 2}}$$

$$\lim_{x \rightarrow -\infty} \frac{x^2 - (x^2 + 2)}{x - \sqrt{x^2 + 2}}$$

$$\lim_{x \rightarrow -\infty} \frac{\cancel{x^2} - \cancel{x^2} - 2}{x - \sqrt{x^2 + 2}}$$

$$= \frac{-2}{-\infty - \sqrt{\infty}} = \frac{-2}{-\infty - \infty} = \frac{-2}{-\infty} = 0$$

Q # 2

$$f(x) = \begin{cases} \frac{x^2 + x - 6}{x + 3} & x < 0 \\ \frac{x + 1}{(x - 1)^2} & x \geq 0 \end{cases}$$

at  $x = 0$

$$f(0) = \frac{0 + 1}{(0 - 1)^2} = \frac{1}{1} = 1$$

$$\lim_{x \rightarrow 0^+} \frac{x + 1}{(x - 1)^2} = \frac{1}{(-1)^2} = 1$$

$$\lim_{x \rightarrow 0^-} \frac{x^2 + x - 6}{x + 3} = \frac{0 + 0 - 6}{0 + 3} = \frac{-6}{3} = -2$$

$\therefore$  at  $x = 0 \rightarrow$  jump discontinuity

$$(\because \lim_{x \rightarrow 0^+} f(x) \neq \lim_{x \rightarrow 0^-} f(x))$$

at  $x = -3$

$f(-3)$  undefined

$$\lim_{x \rightarrow -3} \frac{x^2 + x - 6}{x + 3} = \lim_{x \rightarrow -3} \frac{(x - 2)(x + 3)}{x + 3} = -5$$

at  $x = -3 \rightarrow$  removable discontinuity

$$(\because f(-3) \neq \lim_{x \rightarrow -3} f(x))$$

at  $x = 1$

$f(1)$  undefined

$$\lim_{x \rightarrow 1} \frac{x + 1}{(x - 1)^2} = \frac{2}{0} = \infty$$

$$\lim_{x \rightarrow 1^+} \frac{x + 1}{(x - 1)^2} = \frac{2}{+0} = \infty$$

$$\lim_{x \rightarrow 1^-} \frac{x + 1}{(x - 1)^2} = \frac{2}{-0} = \infty$$

$\therefore$  at  $x = 1 \rightarrow$  infinite discontinuity

Q #3

$$f(x) = \frac{2x^2 + 3}{x\sqrt{x^2 + 4}}$$

HA? .  
V.A?

HA

$$\lim_{x \rightarrow \pm\infty} \frac{2x^2 + 3}{x\sqrt{x^2 + 4}}$$

$$\lim_{x \rightarrow \pm\infty} \frac{2x^2 + 3}{x\sqrt{x^2(1 + \frac{4}{x^2})}} = \lim_{x \rightarrow \pm\infty} \frac{2x^2 + 3}{x|x|\sqrt{1 + \frac{4}{x^2}}}$$

$$\lim_{x \rightarrow \infty} \frac{2x^2 + 3}{x^2 \sqrt{1 + \frac{4}{x^2}}}$$

$$\lim_{x \rightarrow \infty} \frac{\cancel{x^2} (2 + \frac{3}{x^2})}{\cancel{x^2} \sqrt{1 + \frac{4}{x^2}}}$$

$$= \frac{2 + 0}{\sqrt{1 + 0}} = \textcircled{2}$$

$$\lim_{x \rightarrow -\infty} \frac{2x^2 + 3}{-x^2 \sqrt{1 + \frac{4}{x^2}}}$$

$$\lim_{x \rightarrow -\infty} \frac{\cancel{x^2} (2 + \frac{3}{x^2})}{-\cancel{x^2} \sqrt{1 + \frac{4}{x^2}}}$$

$$= \frac{2 + 0}{-\sqrt{1 + 0}} = \textcircled{-2}$$

$\therefore y = 2, y = -2$  are HA's

V.A

$$\begin{aligned} x\sqrt{x^2 + 4} &= 0 \\ x &= 0 & x^2 + 4 &= 0 \\ & & x^2 &= -4 \end{aligned}$$

at  $x = 0$

None

$$\lim_{x \rightarrow 0^+} \frac{2x^2 + 3}{x\sqrt{x^2 + 4}}$$

$$\underset{\substack{\nearrow 0 + 0.1 \\ \searrow 0.1}}{=} \frac{3}{+0} = +\infty$$

$$\lim_{x \rightarrow 0^-} \frac{2x^2 + 3}{x\sqrt{x^2 + 4}}$$

$$\underset{\substack{\nearrow 0 - 0.1 \\ \searrow -0.1}}{=} \frac{3}{-0} = -\infty$$

$\therefore x = 0$  is V.A.

Q #4  $f(x) = \ln(2x+3)$

a)  $f^{-1}(x)$

b)  $D_f - R_f$

a)  $y = \ln(2x+3)$

$e^y = e^{\ln(2x+3)}$

$e^y = 2x+3$

$e^y - 3 = 2x$

$x = \frac{1}{2}(e^y - 3) \Rightarrow \therefore f^{-1}(x) = \frac{1}{2}(e^x - 3)$

b) Domain of  $f(x)$

$2x+3 > 0$

$2x > -3$

$x > -\frac{3}{2}$

$\Rightarrow D_f = (-\frac{3}{2}, \infty)$

Domain of  $f^{-1}(x) = \mathbb{R}$

$\Rightarrow \therefore R_f = \mathbb{R}$

Q #5

$$(5 - 2|x-4|) \leq f(x) + 2 \leq (x+1)$$

$$\lim_{x \rightarrow 4} f(x) ?$$



$$5 - 2|x-4| - 2 \leq f(x) \leq x+1 - 2$$

$$3 - 2|x-4| \leq f(x) \leq x-1$$

$$\lim_{x \rightarrow 4} (3 - 2|x-4|) \quad \bigg| \quad \lim_{x \rightarrow 4} (x-1)$$

$$= 3 - 0$$

$$= 4 - 1$$

$$= \textcircled{3}$$

$$= \textcircled{3}$$

∴ By Sandwich Thm

$$\lim_{x \rightarrow 4} f(x) = 3$$

Q #7

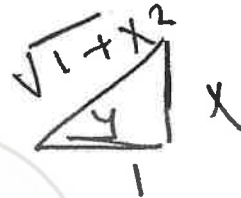
$$a) \sin(\tan^{-1} x) = \frac{x}{\sqrt{1+x^2}}$$

L.H.S

$$y = \tan^{-1} x$$

$$\tan y = \tan \tan^{-1} x$$

$$\tan y = x$$



$$\sin(\tan^{-1} x)$$

$$\sin y = \frac{x}{\sqrt{1+x^2}} =$$

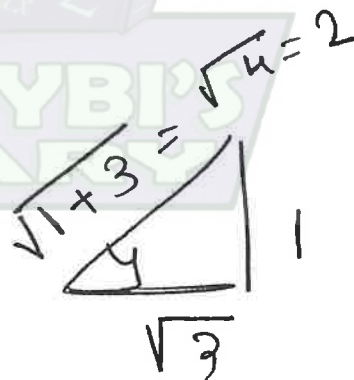
R.H.S

$$b) \sin(\tan^{-1} (\frac{1}{\sqrt{3}}))$$

$$y = \tan^{-1} \frac{1}{\sqrt{3}}$$

$$\tan y = \tan \tan^{-1} \frac{1}{\sqrt{3}}$$

$$\tan y = \frac{1}{\sqrt{3}}$$



OR  
اگر اکوٹ

①  
1/3

$$\rightarrow \sin(\tan^{-1} (\frac{1}{\sqrt{3}}))$$

$$\sin y = \frac{1}{2}$$

#

Calculators and Mobile Phones are not allowed.

Q1. [10+10=20 pts.] Use the graph of  $y = f(x)$  shown below to:

- a) Find the following limits:  
 i)  $\lim_{x \rightarrow -2^-} f(x)$ ;    ii)  $\lim_{x \rightarrow -2^+} f(x)$ ;    iii)  $\lim_{x \rightarrow 1^-} f(x)$ ;    iv)  $\lim_{x \rightarrow 1^+} f(x)$ ;    v)  $\lim_{x \rightarrow 4^-} f(x)$   
 b) Find and classify the discontinuities of  $f$  as removable, infinite or jump.



Figure 1: The graph of  $y = f(x)$ .

Q2. [5+10+5=20 pts.] Let  $f(x) = 4 + e^{(x-3)}$

- a) Find the domain of  $f$ .  
 b) Find  $f^{-1}(x)$ .  
 c) Find the range of  $f$ .

Q3. [20 pts.] Let  $f(x) = \frac{x\sqrt{x^2+1}}{x^2-1}$ . Find the vertical and horizontal asymptotes for the graph of  $f$ , (if any).

Q4. [5+5=10 pts.] Evaluate each of the following limits, if it exists:

i)  $\lim_{x \rightarrow 2} \frac{x^2 - 4}{x^3 - 8}$

ii)  $\lim_{x \rightarrow 1} (x-1)^4 \cos\left(\frac{2}{x-1}\right)$

Q5. [10 pts.] Find a positive value for the constant  $B$ , if any, that makes the function  $f$  continuous at  $x = 2$ , where

$$f(x) = \begin{cases} \frac{8-2x^2}{2-x} & \text{if } x < 2 \\ \frac{B^2-1}{B^2+1} & \text{if } x \geq 2 \end{cases}$$

Q6. [5+5=10 pts.]

- a) Prove that:  $\sin(\cos^{-1} x) = \sqrt{1-x^2}$ .  
 b) Find the exact value of the expression:  $\sin\left(\cos^{-1}\left(\frac{1}{2}\right)\right)$

Q7. [10 pts.] Use the Intermediate Value Theorem to show that the equation:  $3^x + x - 2 = 0$  has at least one real solution.

Q#1

a) i)  $\lim_{x \rightarrow -2^-} f(x) = 3$

ii)  $\lim_{x \rightarrow -2^+} f(x) = 3$

(ii)  $\lim_{x \rightarrow 1^-} f(x) = -3$

(v)  $\lim_{x \rightarrow 1^+} f(x) = 4$

v)  $\lim_{x \rightarrow 4^-} f(x) = -\infty$

يفرق الاتجاهين :  
يفرق الصورة : -2

b) at  $x=1 \rightarrow$  jump disc.

at  $x=4 \rightarrow$  infinite disc.

at  $x=-2 \rightarrow$  Removable disc.

Q# 2

$$f(x) = 4 + e^{x-3}$$

(a) Domain of  $f(x)$ :

$$\Rightarrow D_f = \mathbb{R}$$

(b)  $y = 4 + e^{x-3}$

$$y - 4 = e^{x-3}$$

$$\ln(y-4) = \ln e^{x-3}$$

$$\ln(y-4) = x-3$$

$$x = \ln(y-4) + 3$$

$$\Rightarrow f^{-1}(x) = \ln(x-4) + 3$$

(c) Domain of  $f^{-1}$ :  $x-4 > 0$

$$x > 4 \Rightarrow (4, \infty)$$

$$\Rightarrow \text{Range of } f = (4, \infty)$$

Q#3

$$f(x) = \frac{x \sqrt{x^2 + 1}}{x^2 - 1}$$

HA

$$\lim_{x \rightarrow \pm \infty} \frac{x \sqrt{x^2 + 1}}{x^2 - 1} = \frac{\pm \infty}{\infty}$$

$$\lim_{x \rightarrow \pm \infty} \frac{x \sqrt{x^2 (1 + \frac{1}{x^2})}}{x^2 - 1}$$

$$\lim_{x \rightarrow \pm \infty} \frac{x |x| \sqrt{1 + \frac{1}{x^2}}}{x^2 - 1}$$

$$\lim_{x \rightarrow \infty} \frac{x \cdot x \sqrt{1 + \frac{1}{x^2}}}{x^2 - 1}$$

$$\lim_{x \rightarrow \infty} \frac{\cancel{x^2} \sqrt{1 + \frac{1}{x^2}}}{\cancel{x^2} (1 - \frac{1}{x^2})}$$

$$= \frac{\sqrt{1+0}}{1-0} = \frac{1}{1} = \textcircled{1}$$

$$\lim_{x \rightarrow -\infty} \frac{x(-x) \sqrt{1 + \frac{1}{x^2}}}{x^2 - 1}$$

$$\lim_{x \rightarrow -\infty} \frac{\cancel{x^2} \sqrt{1 + \frac{1}{x^2}}}{\cancel{x^2} (1 - \frac{1}{x^2})}$$

$$= -\frac{\sqrt{1+0}}{1-0} = -\frac{1}{1} = \textcircled{-1}$$

$y = 1$   $y = -1$  are HAs

$\Rightarrow$

النتيجة

V.A

$$x^2 - 1 = 0$$

$$x^2 = 1$$

$$x = 1$$

$$x = -1$$

at  $x = 1$

$$\lim_{x \rightarrow 1^+} \frac{x \sqrt{x^2 + 1}}{(x-1)(x+1)}$$

$$1 + 0.1 = 1.1$$

$$= \frac{\sqrt{2}}{0^+} = +\infty$$

$\therefore x = 1$  is V.A

$$\lim_{x \rightarrow 1^-} \frac{x \sqrt{x^2 + 1}}{(x-1)(x+1)}$$

$$1 - 0.1 = 0.9$$

$$= \frac{\sqrt{2}}{0^-} = -\infty$$

at  $x = -1$

$$\lim_{x \rightarrow -1^+} \frac{x \sqrt{x^2 + 1}}{(x-1)(x+1)}$$

$$-1 + 0.1 = -0.9$$

$$= \frac{-\sqrt{2}}{0^-} = +\infty$$

$$\lim_{x \rightarrow -1^-} \frac{x \sqrt{x^2 + 1}}{(x-1)(x+1)}$$

$$-1 - 0.1 = -1.1$$

$$= \frac{-\sqrt{2}}{0^+} = -\infty$$

$\therefore x = -1$  is V.A

Q#4

$$\begin{aligned} \text{i) } \lim_{x \rightarrow 2} \frac{x^2 - 4}{x^3 - 8} & \quad \frac{0}{0} \\ \lim_{x \rightarrow 2} \frac{\cancel{(x-2)}(x+2)}{\cancel{(x-2)}(x^2 + 2x + 4)} &= \frac{4}{4+4+4} \\ &= \frac{4}{12} = \boxed{\frac{1}{3}} \end{aligned}$$

$$\begin{aligned} \text{ii) } \lim_{x \rightarrow 1} (x-1)^4 \cos\left(\frac{2}{x-1}\right) & \quad 0 \cos \infty \\ -1 \leq \cos \theta \leq 1 \quad \forall \theta \\ -1 \leq \cos\left(\frac{2}{x-1}\right) \leq 1 \\ \infty (x-1)^4 \neq 0 \\ -(x-1)^4 \leq (x-1)^4 \cos\left(\frac{2}{x-1}\right) \leq (x-1)^4 \\ \lim_{x \rightarrow 1} -(x-1)^4 = 0 \quad \lim_{x \rightarrow 1} (x-1)^4 = 0 \end{aligned}$$

By Sandwich Thm

$$\lim_{x \rightarrow 1} (x-1)^4 \cos\left(\frac{2}{x-1}\right) = 0$$

Q#5

$$f(x) = \begin{cases} \frac{8-2x^2}{2-x} & , x < 2 \\ \frac{B^{2x}-1}{B^x+1} & , x \neq 2 \end{cases}$$

$$f(2) = \frac{B^4-1}{B^2+1} = \frac{(B^2-1)(B^2+1)}{B^2+1} = \boxed{B^2-1}$$

$$\lim_{x \rightarrow 2^+} \frac{B^{2x}-1}{B^x+1} = \frac{B^4-1}{B^2+1} = \frac{(B^2-1)(B^2+1)}{B^2+1} = \boxed{B^2-1}$$

$$\lim_{x \rightarrow 2^-} \frac{8-2x^2}{2-x} = \frac{0}{0}$$

$$\lim_{x \rightarrow 2^-} \frac{2(4-x^2)}{2-x}$$

$$\lim_{x \rightarrow 2^-} \frac{2(2-x)(2+x)}{2-x} = 2(4) = \boxed{8}$$

∴  $f$  is continuous at  $x=2$

$$\therefore f(2) = \lim_{x \rightarrow 2^+} f(x) = \lim_{x \rightarrow 2^-} f(x)$$

$$\underline{B^2-1 = B^2-1 = 8}$$

$$B^2 = 9$$

$$\boxed{B = \pm 3} \xrightarrow[\text{موجب}]{\text{مثالي}} \boxed{B = 3}$$

Q #6

@ Prove :  $\sin(\cos^{-1} x) = \sqrt{1-x^2}$

$$\text{LHS} = \sin(\cos^{-1} x)$$

$$y = \cos^{-1} x$$

$$\cos y = \cos \cos^{-1} x$$

$$\cos y = x$$



$$\Rightarrow \sin(\cos^{-1} x)$$

$$\sin y = \frac{\sqrt{1-x^2}}{1} = \sqrt{1-x^2} = \text{RHS}$$

(b)  $\sin(\cos^{-1} \frac{1}{2})$

at  $x = \frac{1}{2}$

(( using (a) ))

فالدای اوله  
من  
اول

استفید  
من  
ا

$$\sin(\cos^{-1} \frac{1}{2}) = \sqrt{1 - (\frac{1}{2})^2}$$

$$= \sqrt{1 - \frac{1}{4}}$$

$$= \sqrt{\frac{4}{4} - \frac{1}{4}}$$

$$= \sqrt{\frac{3}{4}}$$

$$= \frac{\sqrt{3}}{2}$$

Q#7

$$3^x + x - 2 = 0$$

Let  $f(x) = 3^x + x - 2$

①  $f(0) = 3^0 + 0 - 2 = 1 - 2 = -1 < 0$

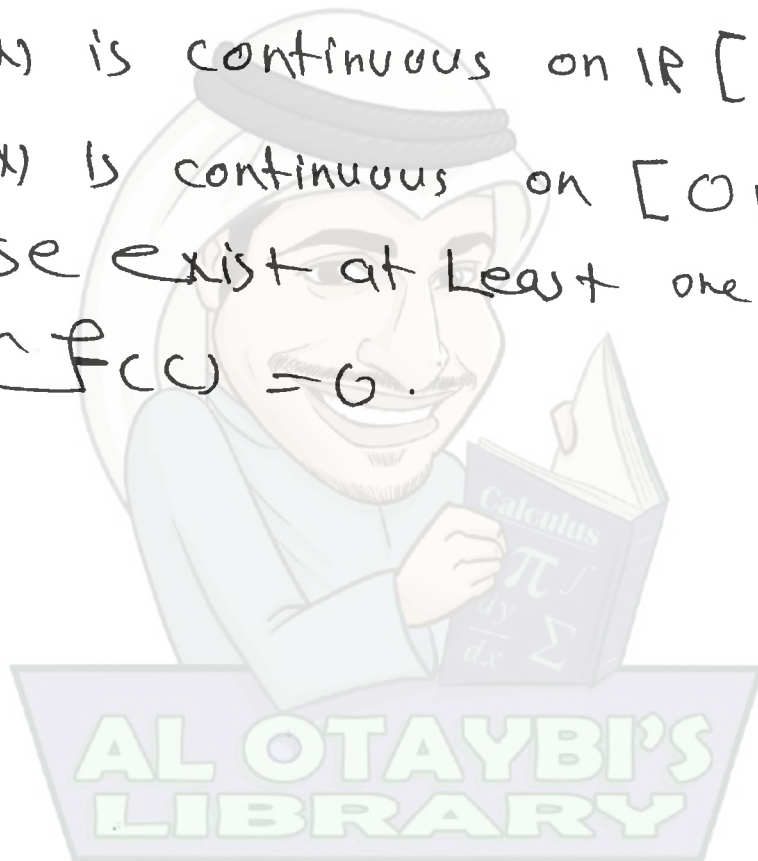
②  $f(1) = 3^1 + 1 - 2 = 2 > 0$

From ① ②  $\rightarrow$  By I.V.T

Since  $f(x)$  is continuous on  $\mathbb{R}$  [ $3^x$ : exponential fu,  
then  $f(x)$  is continuous on  $[0, 1]$  [ $x-2$ : polynomial

then there exist at least one  $c \in (0, 1)$

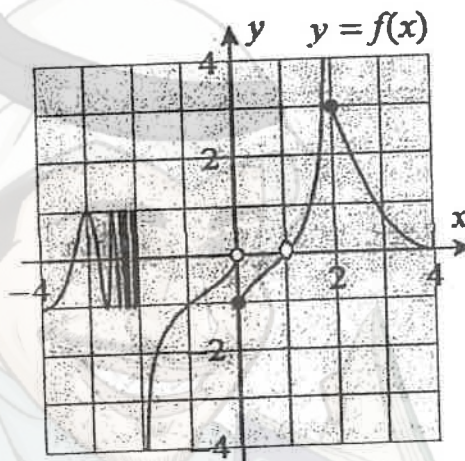
s.t.  $f(c) = 0$ .



Answer all questions. Calculators and mobile phones are not allowed.

Q1. [10+10=20 pts.] Use the graph of  $y = f(x)$  shown below to:

- a) Find the following limits:  
 i)  $\lim_{x \rightarrow 0^-} f(x)$ ;    ii)  $\lim_{x \rightarrow 0^+} f(x)$ ;    iii)  $\lim_{x \rightarrow 1} f(x)$ ;    iv)  $\lim_{x \rightarrow 2^+} f(x)$ ;    v)  $\lim_{x \rightarrow -2^+} f(x)$ .  
 b) Find and classify the discontinuities of  $f$  as removable, infinite or jump.



Q2. [5+5+10=20 pts.] Let  $f(x) = 2^x + \ln(x)$  and  $g(x) = \frac{x-1}{x+1}$ .

- (a) Find  $\lim_{x \rightarrow 1} (f(x) + 3g(x))$ .  
 (b) If  $f(1) = 2$ , find  $f^{-1}(2)$ .  
 (c) Find  $g^{-1}(x)$ .

Q3. [10+10=20 pts.] Evaluate each of the following limits, if it exists:

(a)  $\lim_{x \rightarrow 0} \frac{xe^x - x^2}{x \sin^2(x)}$

(b)  $\lim_{x \rightarrow 1} \cos^{-1} \left( \frac{x-1}{x^2-1} \right)$

Q4. [20 pts.] Let  $f(x) = \frac{x\sqrt{x^2+2}}{1-x^2}$ .

Find the vertical and horizontal asymptotes for the graph of  $f$ , if any.

Q5. [10 pts.] Let  $f(x) = \frac{2}{\sqrt{x}}$ . Use the definition of the derivative to find  $f'(4)$ .

Q6. [10 pts.] Let  $f(x) = 2 + x \sin(x)$  and  $g(x) = x^2$ . Use the Intermediate Value Theorem to show that the graphs of  $f$  and  $g$  intersect at least once.

Q #1

i)  $\lim_{x \rightarrow 0^-} f(x) = 0$

ii)  $\lim_{x \rightarrow 0^+} f(x) = -1$

iii)  $\lim_{x \rightarrow 1} f(x) = 0$

iv)  $\lim_{x \rightarrow 2^+} f(x) = 3$

v)  $\lim_{x \rightarrow -2^+} f(x) = -\infty$

يفرق الاتجاهين  $-2$  و  $2$   
يفرق الصورة  $1$  و  $2$

b) at  $x = -2 \rightarrow$  infinite disc.

at  $x = 0 \rightarrow$  jump disc.

at  $x = 2 \rightarrow$  infinite disc.

at  $x = 1 \rightarrow$  Removable disc.

Q # 2  $f(x) = 2^x + \ln x$  ,  $g(x) = \frac{x-1}{x+1}$

(a) Find  $\lim_{x \rightarrow 1} (f(x) + 3g(x))$

$$\lim_{x \rightarrow 1} f(x) = \lim_{x \rightarrow 1} (2^x + \ln x) = 2^1 + \ln 1 = 2 + 0 = \textcircled{2}$$

$$\lim_{x \rightarrow 1} g(x) = \lim_{x \rightarrow 1} \frac{x-1}{x+1} = \frac{0}{2} = \textcircled{0}$$

$$\therefore \lim_{x \rightarrow 1} (f(x) + 3g(x)) = (2 + 3(0)) = 2 + 0 = \textcircled{2}$$

(b) If  $f(1) = 2$  Find  $f^{-1}(2)$ .

$$\therefore f(1) = 2 \rightarrow \therefore f^{-1}(2) = 1$$

(c)  $y = \frac{x-1}{x+1}$

$$y(x+1) = x-1$$

$$yx + y = x - 1$$

$$yx - x = -1 - y$$

$$x(y-1) = -1-y$$

$$x = \frac{-1-y}{y-1}$$

$$\therefore f^{-1}(x) = \frac{-1-x}{x-1}$$

Q#3

(a)  $\lim_{x \rightarrow 0} \frac{x e^x - x^2}{x \sin^2 x}$   $\frac{0}{0}$

$\lim_{x \rightarrow 0} \frac{x(e^x - x)}{x \sin^2 x}$   
 $\frac{x-0}{0-x}$

$$\lim_{x \rightarrow 0} \frac{e^x - x}{\sin^2 x} = \frac{1-0}{0} = \frac{1}{0} = \infty$$

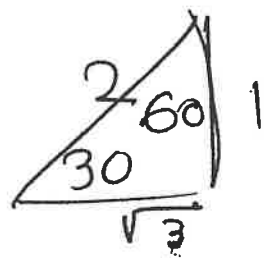
(b)  $\lim_{x \rightarrow 1} \cos^{-1}\left(\frac{x-1}{x^2-1}\right)$   $\cos^{-1} \frac{0}{0}$

$\frac{x-1}{1-x}$

$\lim_{x \rightarrow 1} \cos^{-1}\left(\frac{x-1}{(x-1)(x+1)}\right)$

$$= \cos^{-1}\left(\frac{1}{2}\right)$$

$$= \frac{\pi}{3}$$



Q#4

$$f(x) = \frac{x\sqrt{x^2+2}}{1-x^2}$$

HA

$$\lim_{x \rightarrow \pm\infty} \frac{x\sqrt{x^2+2}}{1-x^2}$$

$$\lim_{x \rightarrow \pm\infty} \frac{x\sqrt{x^2(1+\frac{2}{x^2})}}{1-x^2}$$

$$\lim_{x \rightarrow \pm\infty} \frac{x|x|\sqrt{1+\frac{2}{x^2}}}{1-x^2}$$

$$\lim_{x \rightarrow \infty} \frac{x(x)\sqrt{1+\frac{2}{x^2}}}{1-x^2}$$

$$\lim_{x \rightarrow -\infty} \frac{x(-x)\sqrt{1+\frac{2}{x^2}}}{1-x^2}$$

$$\lim_{x \rightarrow \infty} \frac{\cancel{x^2}\sqrt{1+\frac{2}{x^2}}}{\cancel{x^2}(\frac{1}{x^2}-1)}$$

$$\lim_{x \rightarrow -\infty} \frac{-\cancel{x^2}\sqrt{1+\frac{2}{x^2}}}{\cancel{x^2}(\frac{1}{x^2}-1)}$$

$$= \frac{\sqrt{1+0}}{0-1} = \frac{1}{-1} = -1 \quad \quad \quad = \frac{-\sqrt{1+0}}{0-1} = \frac{-1}{-1} = 1$$

$y = -1$   $y = 1$  are H.A's.

$\Rightarrow$

$$\text{V.A } 1-x^2=0$$

$$x^2=1$$

$$x=1$$

$$x=-1$$

at  $x=1$

$$\lim_{x \rightarrow 1^+} \frac{x\sqrt{x^2+2}}{(1-x)(1+x)}$$

$$\frac{\sqrt{3}}{0^-} = -\infty$$

$$1 + 0.1 = 1.1$$

$$\lim_{x \rightarrow 1^-} \frac{x\sqrt{x^2+2}}{(1-x)(1+x)}$$

$$\frac{\sqrt{3}}{0^+} = +\infty$$

$$1 - 0.1 = 0.9$$

$x=1$  is V.A

at  $x=-1$

$$\lim_{x \rightarrow -1^+} \frac{x\sqrt{x^2+2}}{(1-x)(1+x)}$$

$$\frac{-\sqrt{3}}{0^+} = -\infty$$

$$-1 + 0.1 = -0.9$$

$$\lim_{x \rightarrow -1^-} \frac{x\sqrt{x^2+2}}{(1-x)(1+x)}$$

$$\frac{-\sqrt{3}}{0^-} = +\infty$$

$$-1 - 0.1 = -1.1$$

$x=-1$  is V.A

Q#5

$$f(x) = \frac{2}{\sqrt{x}}$$

Find  $f'(4)$

$$f'(a) = \lim_{x \rightarrow a} \frac{f(x) - f(a)}{x - a}$$

$$= \lim_{x \rightarrow 4} \frac{\frac{2}{\sqrt{x}} - f(4)}{x - 4} \quad \frac{2}{\sqrt{4}} = \frac{2}{2} = 1$$

$$= \lim_{x \rightarrow 4} \frac{\frac{2}{\sqrt{x}} - 1}{x - 4} \quad \frac{0}{0}$$

$$\begin{matrix} x-4 \\ 4-x \end{matrix}$$

ملوك ← ملوك

$$= \lim_{x \rightarrow 4} \frac{\frac{2}{\sqrt{x}} - \frac{\sqrt{x}}{\sqrt{x}}}{x - 4}$$

$$= \lim_{x \rightarrow 4} \frac{\frac{2 - \sqrt{x}}{\sqrt{x}}}{x - 4}$$

$$= \lim_{x \rightarrow 4} \frac{2 - \sqrt{x}}{\sqrt{x} (x - 4)} \quad * \frac{2 + \sqrt{x}}{2 + \sqrt{x}}$$

$$= \lim_{x \rightarrow 4} \frac{4 - x}{\sqrt{x} (x - 4) (2 + \sqrt{x})}$$

$$= \lim_{x \rightarrow 4} \frac{-\cancel{(x-4)}}{\sqrt{x} \cancel{(x-4)} (2 + \sqrt{x})}$$

$$= \frac{-1}{2(2+2)} = \frac{-1}{2(4)} = \frac{-1}{8}$$

$$f(x) = 2x^{-\frac{1}{2}}$$

$$f'(x) = -x^{-\frac{3}{2}}$$

$$= \frac{-1}{x^{\frac{3}{2}}}$$

$$f'(4) = \frac{-1}{(4)^{\frac{3}{2}}}$$

$$= \frac{-1}{2^3} = \frac{-1}{8}$$

Q #6       $f(x) = 2 + x \sin x$        $g(x) = x^2$

Let  $h(x) = f(x) - g(x)$

$$h(x) = 2 + x \sin x - x^2$$

①  $h(0) = 2 + 0 - 0 = 2 > 0$

$$\begin{aligned} h\left(\frac{\pi}{2}\right) &= 2 + \frac{\pi}{2} \sin \frac{\pi}{2} - \left(\frac{\pi}{2}\right)^2 \\ &= 2 + \frac{\pi}{2}(1) - \frac{\pi^2}{4} > 0 \end{aligned}$$

②  $h(\pi) = 2 + \pi \sin \pi - \pi^2$   
 $= 2 + 0 - \pi^2$   
 $= 2 - \pi^2 < 0$

From ① ②  $\rightarrow$  By I.V.T

since  $h(x)$  is continuous on  $\mathbb{R}$  [  $\sin x$ : Trigon  
then  $h(x)$  is continuous on  $[0, \pi]$   $x, x^2$ : poly.

then there exist at least one  $c \in (0, \pi)$   
s.t.  $h(c) = 0$

$$f(c) - g(c) = 0$$

$$f(c) = g(c)$$

$\therefore f$  and  $g$  intersect.